1D Regional Sampling
Approach 2

1 Hoeffding Approach

We have a function f that is differentiable and has a bounded first derivative, and we want to estimate $\int_a^b f(x)dx$.

1. f differentiable, hence locally Lipschitz.
2. By Mean Value Theorem, the Lipschitz constant of f on some interval $[x, y]$ is $L \leq \max_{z\in[x,y]} f'(z)$.
3. By Extreme Value Theorem, f is also bounded on $[a, b]$.
4. We split $[a, b] = \bigcup_{i=0}^{m-1} [x_i, x_{i+1}]$ where $x_0 = a$ and $x_m = b$.
5. Denote total budget as N, and for region $[x_k, x_{k+1}]$ we sample n_k samples.
6. We estimate

$$\begin{aligned}\int_{x_k}^{x_{k+1}} f(x)dx = \int_{x_t}^{x_{k+1}} f(x)(x_{k+1} - x_k)\frac{1}{x_{k+1} - x_k}dx &= (x_{k+1} - x_k)\mathbb{E}_{X\sim \mathrm{Unif}[x_k,x_{k+1}]}[f(X)] \\ &\approx \frac{x_{k+1} - x_k}{n_k}\sum_{i=1}^{n_k} f(X_i) \qquad X_i \sim \mathrm{Unif}[x_k, x_{k+1}]\end{aligned}$$

for $0 \leq k \leq m - 1$. Therefore, we get

$$\begin{aligned}I := \int_a^b f(x)dx &= \sum_{k=0}^{m-1} \int_{x_k}^{x_{k+1}} f(x)dx \\ &= \sum_{k=0}^{m-1} (x_{k+1} - x_k)\mathbb{E}_{X\sim[x_k,x_{k+1}]}[f(X)] \\ &\approx \sum_{k=0}^{m-1} \left(\frac{x_{k+1} - x_k}{n_k}\sum_{i=1}^{n_k} f(X_i \sim [x_k, x_{k+1}])\right) =: \hat{I}\end{aligned}$$

7. We have the following concentration bound

$$\begin{aligned}\mathbb{P}(|\hat{I} - I| > \varepsilon) &= \mathbb{P}\left(\left|\sum_{k=0}^{m-1}\left(\frac{x_{k+1} - x_k}{n_k}\sum_{i=1}^{n_k} f(X_i \sim [x_k, x_{k+1}])\right) - \sum_{k=0}^{m-1}(x_{k+1} - x_k)\mathbb{E}_{X\sim[x_k,x_{k+1}]}[f(X)]\right| > \varepsilon\right) \\ &= \mathbb{P}\left(\left|\sum_{k=0}^{m-1}\sum_{i=1}^{n_k}\left\{\frac{x_{k+1} - x_k}{n_k}f(X_i \sim [x_k, x_{k+1}]) - \mathbb{E}_{X\sim[x_k,x_{k+1}]}\left[\frac{x_{k+1} - x_k}{n_k}f(X)\right]\right\}\right| \geq \varepsilon\right)\end{aligned}$$

If we denote $Z_{k,i} = \frac{x_{k+1}-x_k}{n_k} f(X_i \sim [x_k, x_{k+1}])$, then we have

$$\mathbb{P}(|\hat{I} - I| \geq \varepsilon) = \mathbb{P}\left(\left|\sum_{k=0}^{m-1}\sum_{i=1}^{n_k}\{Z_{k,i} - \mathbb{E}[Z_{k,i}]\}\right| \geq \varepsilon\right) \tag{1}$$

for any $0 \le k \le m-1, 1 \le i \le n_k$ we have $X_i \sim \text{Unif}[x_k, x_{k+1}]$ is bounded, and thus $f(X_i)$ is bounded $\implies f(X_i)$ is subgaussian $\implies Z_{k,i}$ is subgaussian. Now we apply Hoeffding to (1) to obtain

$$\mathbb{P}(|\hat{I} - I| \ge \varepsilon) = \mathbb{P}\left(\left|\sum_{k=0}^{m-1}\sum_{i=1}^{n_k}\{Z_{k,i} - \mathbb{E}[Z_{k,i}]\}\right| \ge \varepsilon\right) \le 2\exp\left(-\frac{c\varepsilon^2}{\sum\limits_{k=0}^{m-1}\sum\limits_{i=1}^{n_k}\|Z_{k,i} - \mathbb{E}[Z_{k,i}]\|_{\psi_2}^2}\right) \tag{2}$$

From the centering lemma, we have

$$\|Z_{k,i} - \mathbb{E}[Z_{k,i}]\|_{\psi_2} \le C\|Z_{k,i}\|_{\psi_2} = C\left(\frac{x_{k+1} - x_k}{n_k}\right)^2 \|f(X \sim [x_k, x_{k+1}])\|_{\psi_2}^2$$

for some absolute constant C. Substituting into (2) we obtain

$$\mathbb{P}(|\hat{I} - I| \ge \varepsilon) \le 2\exp\left(-\frac{\tilde{c}\varepsilon^2}{\sum\limits_{k=0}^{m-1}\frac{(x_{k+1}-x_k)^2}{n_k}\|f(X \sim [x_k, x_{k+1}])\|_{\psi_2}^2}\right) \tag{3}$$

where $\tilde{c} := c/C$ is absolute constant.

8. Following from point 2., we denote the min and max that f takes on $[x_k, x_{k+1}]$ as m_k, M_k respectively, then we obtain another classic Hoeffding concentration bound from (2)

$$\mathbb{P}(|\hat{I} - I| \ge \varepsilon) \le 2\exp\left(-\frac{2\varepsilon^2}{\sum\limits_{k=0}^{m-1}\frac{(x_{k+1}-x_k)^2}{n_k}(M_k - m_k)^2}\right) \tag{4}$$

Optimal allocation (single split)

Consider we have $m = 2$ (i.e only two regions from a single split), we obtain from (3)

$$\mathbb{P}(|\hat{I} - I| \ge \varepsilon) \le 2\exp\left(-\frac{2\varepsilon^2}{\frac{(x-a)^2}{n_1}(M_1 - m_1)^2 + \frac{(b-x)^2}{N-n_1}(M_2 - m_2)^2}\right) \tag{5}$$

where $x \in [a, b]$ is the some fixed splitting point and n_1 is the number of sample allocated to region $[a, x]$. Minimising RHS in (4) is equivalent to minimising

$$g(n_1) = \frac{(x-a)^2}{n_1}(M_1 - m_1)^2 + \frac{(b-x)^2}{N-n_1}(M_2 - m_2)^2$$

First we note that g is strictly convex on $[0, N]$, thus the stationary point is global minimiser. Differentiating g gives

$$\frac{dg}{dn_1} = -\frac{(x-a)^2}{n_1^2}(M_1 - m_1)^2 + \frac{(b-x)^2}{(N-n_1)^2}(M_2 - m_2)^2 = 0 \implies n_1^* = N\frac{(x-a)(M_1 - m_1)}{(x-a)(M_1 - m_1) + (b-x)(M_2 - m_2)}$$

or more precisely, we should take

$$\boxed{n_1^* = \text{round}\left(N\frac{(x-a)(M_1 - m_1)}{(x-a)(M_1 - m_1) + (b-x)(M_2 - m_2)}\right)} \tag{6}$$

or to keep it scale-less, we write in terms of proportions to allocate to n_1, which is

$$\boxed{\frac{n_1^*}{N} = \text{round}\left(\frac{(x-a)(M_1 - m_1)}{(x-a)(M_1 - m_1) + (b-x)(M_2 - m_2)}\right)} \tag{7}$$

Optimal Spliting

We can use the n_1^* determined in (5) to obtain the concentration optimal splitting point x. Since n_1^* minimises RHS for any fixed splitting, we can just sub (6) into (5) and minimise the RHS wrt x. This is equivalent to minimising $g(n_1^*(x))$, we have

$$\begin{aligned}
g(n_1^*(x)) &= \frac{(x-a)^2(M_1-m_1)^2}{N\frac{(x-a)(M_1-m_1)}{(x-a)(M_1-m_1)+(b-x)(M_2-m_2)}} + \frac{(b-x)^2(M_2-m_2)^2}{N\frac{(b-x)(M_2-m_2)}{(x-a)(M_1-m_1)+(b-x)(M_2-m_2)}} \\
&= \frac{1}{N}[(x-a)(M_1-m_1)+(b-x)(M_2-m_2)][(x-a)(M_1-m_1)+(b-x)(M_2-m_2)] \\
&= \frac{1}{N}[(x-a)(M_1-m_1)+(b-x)(M_2-m_2)]^2
\end{aligned}$$

thus minimising $g(n_1^*(x))$ is equivalent to minimising the squared term, and since

$$(x-a)(M_1-m_1)+(b-x)(M_2-m_2) \geq 0 \qquad a \leq x \leq b \tag{8}$$

it is equivalent to minimising $(x-a)(M_1-m_1)+(b-x)(M_2-m_2)$. Note that m_1, m_2, M_1 and M_2 depend on x as well. Thus, I wouldn't expect there would be a general minimiser x^*.

Example

Take $f(x) = e^x, [a,b] = [0,10], N = 1000, m = 2$. This particular example simplifies (8) to

$$h(x) = x(e^x - 1) + (10-x)(e^{10} - e^x).$$

simplifying and differentiating

$$\begin{aligned}
h(x) &= xe^x - x + (10-x)e^{10} - (10-x)e^x \\
&= (2x-10)e^x + (10-x)e^{10} - x \\
h'(x) &= \frac{d}{dx}\left[(2x-10)e^x\right] - e^{10} - 1 \\
&= e^x(2x-8) - e^{10} - 1
\end{aligned}$$

Setting $h'(x) = 0$ gives

$$e^x(2x-8) = e^{10} + 1$$

Let $y = 2x - 8$, then

$$ye^{y/2} = (e^{10}+1)e^{-4} \implies y = 2\,W\left(\tfrac{1}{2}(e^{10}+1)e^{-4}\right)$$

where W is the Lambert W function. Therefore unique stationary point is

$$\boxed{x^* = 4 + W\left(\tfrac{1}{2}(e^{10}+1)e^{-4}\right) \approx 7.9366}$$

and it can be shown that h is convex on $[3,10]$ and $x^* \in [3,10]$ thus this is a minimiser. The allocation follows from (6)

$$\boxed{n_1^* = \text{round}\left(\frac{1000(7.94)(e^{7.94}-1)}{7.94(e^{7.94}-1)+(10-7.94)(e^{10}-e^{7.94})}\right) = 360}$$

Greedy Hoeffding Algorithm

Given a function f (assume we can minimse (8) with such f) and we wish to compute $\int_a^b f(x)dx$ using a budget of only N sample points, we follow the following *Greedy Hoeffding algorithm.* Because we are splitting into two regions at each time, we will denote n_1, n_2 as n_l, n_r respectively.

Step 1. **(Split-point optimisation)**

Given a region $[a,b]$ with remaining random-sampling budget n and stored sample set S, define the effective budget

$$n_{\text{eff}} := n + |S|.$$

Draw a finite set of candidate split points

$$\{x_1, \ldots, x_K\} \subset (a, b)$$

uniformly at random.

For each candidate x_k, estimate the extrema on the two subintervals using a deterministic optimisation routine (e.g. Newton's method), initialised from previously stored samples and, if needed, additional random initialisations capped by an exploration budget. Specifically,

$$\widehat{M}_0(x_k) \approx \max_{t \in [a, x_k]} f(t), \quad \widehat{m}_0(x_k) \approx \min_{t \in [a, x_k]} f(t),$$

$$\widehat{M}_1(x_k) \approx \max_{t \in [x_k, b]} f(t), \quad \widehat{m}_1(x_k) \approx \min_{t \in [x_k, b]} f(t).$$

Using these estimates, compute the Hoeffding splitting objective

$$h(x_k) := (x_k - a)\big(\widehat{M}_0(x_k) - \widehat{m}_0(x_k)\big) + (b - x_k)\big(\widehat{M}_1(x_k) - \widehat{m}_1(x_k)\big).$$

Select the split point

$$x^* = \arg\min_{x_k} h(x_k).$$

Step 2. **(Exploration sampling)**

Let $\alpha \in (0, 1)$ denote the exploration fraction. Define the exploration target

$$n_{\text{explore}} := \max\big(1, \text{round}(\alpha\, n_{\text{eff}})\big).$$

If $|S| < n_{\text{explore}}$, draw

$$\min(n_{\text{explore}} - |S|,\ n)$$

additional random samples uniformly from $[a, b]$, evaluate f at those points, add them to S, and decrement the remaining budget n accordingly.

All random initialisation points used during the optimisation in Step 1 are counted toward this exploration budget and are retained as samples.

Step 3. **(Budget allocation)**

Using the estimated ranges at x^*, allocate the remaining random-sampling budget n between the two subregions according to the Hoeffding allocation rule:

$$n_l^* = \text{round}\left(n\, \frac{(x^* - a)\big(\widehat{M}_0(x^*) - \widehat{m}_0(x^*)\big)}{(x^* - a)\big(\widehat{M}_0(x^*) - \widehat{m}_0(x^*)\big) + (b - x^*)\big(\widehat{M}_1(x^*) - \widehat{m}_1(x^*)\big)}\right),$$

and set

$$n_r^* := n - n_l^*.$$

Step 4. **(Recursive splitting)**

Let $\beta \in (0, 1)$ denote the stopping fraction relative to the *global* budget N. Apply Steps 1–3 recursively to the left subregion $[a, x^*]$ with budget n_l^* if

$$n_l^* + |S \cap [a, x^*]| > \beta\, N,$$

and to the right subregion $[x^*, b]$ with budget n_r^* if

$$n_r^* + |S \cap [x^*, b]| > \beta\, N.$$

All stored samples are assigned to their corresponding subregions and reused for subsequent split decisions.

Step 5. **(Termination)**

The algorithm terminates when every region has effective budget

$$n_{\text{eff}} = (\text{remaining budget}) + (\text{stored samples})$$

less than or equal to $\beta\, N$.

2 Bernstein Approach

With similar setups, we can apply Bernstein inequality to (1)

$$\mathbb{P}(|\hat{I}-I| \geq \varepsilon) = \mathbb{P}\left(\left|\sum_{k=0}^{m-1}\sum_{i=1}^{n_k}\{Z_{k,i}-\mathbb{E}[Z_{k,i}]\}\right| \geq \varepsilon\right) \leq 2\exp\left(-\frac{\varepsilon^2/2}{\sum\limits_{k=0}^{m-1}\sum\limits_{i=1}^{n_k}\mathbb{E}[(Z_{k,i}-\mathbb{E}[Z_{k,i}])^2]+B\varepsilon/3}\right) \tag{9}$$

where B is such that

$$|Z_{k,1}-\mathbb{E}[Z_{k,1}]| = \frac{x_{k+1}-x_k}{n_k}\left|f(X_1)-\mathbb{E}_{X\sim[x_k,x_{k+1}]}[f(X)]\right| < B \qquad \text{almost surely}$$

we can take $B = \max\limits_{0\leq k\leq m-1} \frac{x_{k+1}-x_k}{n_k}(M_k - m_k)$. For simplicity, we take

$$B = (b-a)(M-m) \tag{10}$$

where m, M are the maximum and minimum f takes on $[a,b]$ respectively. Note that this might not give us the same optimisation later, but it's easier optimise.

We have for any fixed k, then for any $1 \leq i \leq n_k$

$$\begin{aligned}\mathbb{E}[(Z_{k,i}-\mathbb{E}[Z_{k,i}])^2] &= \mathrm{Var}_{X\sim[x_k,x_{k+1}]}\left(\frac{x_{k+1}-x_k}{n_k}f(X)\right)\\ &= \left(\frac{x_{k+1}-x_k}{n_k}\right)^2 \mathrm{Var}_{X\sim[x_k,x_{k+1}]}(f(X))\end{aligned}$$

rewritting (9) using the shorthand $\mathbb{V}_k(f(X)) = \mathrm{Var}_{X\sim[x_k,x_{k+1}]}(f(X))$ (or simple $\mathbb{V}_k(f)$) for $0 \leq k \leq m-1$

$$\mathbb{P}(|\hat{I}-I| \geq \varepsilon) = \mathbb{P}\left(\left|\sum_{k=0}^{m-1}\sum_{i=1}^{n_k}\{Z_{k,i}-\mathbb{E}[Z_{k,i}]\}\right| \geq \varepsilon\right) \leq 2\exp\left(-\frac{\varepsilon^2/2}{\sum\limits_{k=0}^{m-1}\frac{(x_{k+1}-x_k)^2}{n_k}\mathbb{V}_k(f(X))+B\varepsilon/3}\right) \tag{11}$$

Optimal allocation

Again, consider $m=2$, we have the corresponding (changing n_1 to n_l now)

$$g(n_l) = \frac{(x-a)^2}{n_l}\mathbb{V}_l(f) + \frac{(b-x)^2}{N-n_l}\mathbb{V}_r(f)$$

again this is convex in n_l, which we can easily find the minimiser n_l^* as

$$\boxed{n_l^* = N\frac{(x-a)\cdot \mathrm{sd}_l(f)}{(x-a)\cdot\mathrm{sd}_l(f)+(b-x)\cdot\mathrm{sd}_r(f)}}$$

where $\mathrm{sd}_j(f) = \sqrt{\mathbb{V}_j(f)}$ for $j \in \{r, l\}$.

Optimal Splitting

With $m=2$, similarly, we get minimisation problem of the form in 8

$$(x-a)\cdot\mathrm{sd}_l(f) + (b-x)\cdot\mathrm{sd}_r(f) \qquad a \leq x \leq b$$

Instead of solving this directly, we **solve an approximated version below 12**, first we introduce the following notation: For a sequence of points $\mathcal{S} = (x_1,\ldots,x_n)$ where $x_i \in [a,b]$, we denote

$$L(x;\mathcal{S},(a,b)) := \sum_{i=1}^{n}\mathbb{I}_{\{x_i\in[a,x]\}} \qquad\qquad R(x;\mathcal{S},(a,b)) := \sum_{i=1}^{n}\mathbb{I}_{x_i\in[x,b]}$$

then we estimate $\mathrm{sd}_0(f)$ and $\mathrm{sd}_1(f)$ as

$$\hat{\mathrm{sd}}_l(x;f) = \sqrt{\frac{1}{L(x;\mathcal{S},(a,b))-1}\sum_{x_i\in[0,x]}\left(f(x_i) - \frac{1}{L(x;\mathcal{S},(a,b))}\sum_{x_i\in[0,x]} f(x_i)\right)^2}$$
$$\hat{\mathrm{sd}}_r(x;f) = \sqrt{\frac{1}{R(x;\mathcal{S},(a,b))-1}\sum_{x_i\in[x,b]}\left(f(x_i) - \frac{1}{R(x;\mathcal{S},(a,b))}\sum_{x_i\in[x,b]} f(x_i)\right)^2}$$

hence our modified optimisation problem setup

$$\min_{a\le x\le b}\left\{h(x) = (x-a)\cdot\hat{\mathrm{sd}}_l(f) + (b-x)\cdot\hat{\mathrm{sd}}_r(f)\right\} \tag{12}$$

Note that as x goes from a to b, the terms $\hat{\mathrm{sd}}_l(f), \hat{\mathrm{sd}}_r(f)$ only changes value when x crosses some point x_i in $\mathcal{S}$. Since $(x-a)$ and $(b-x)$ are linear, hence h attains minimum at one of the points in $\mathcal{S}$. Thus,

$$x^* = \arg\min_{x\in\mathcal{S}} h(x)$$

Optimal Allocation

After obtaining x^*, we can also retrieve $\hat{\mathrm{sd}}_l^*(f)$ and $\hat{\mathrm{sd}}_r^*(f)$ (which are the standard deviation of f at each region in the optimal splitting), then the remaining budget for the left and righ region to continue the process are

$$n_l^* = \mathrm{round}\left((N-|\mathcal{S}|)\frac{(x^*-a)\cdot\hat{\mathrm{sd}}_l^*(f)}{h(x^*)}\right)$$
$$n_r^* = (N-|\mathcal{S}|) - n_l^*$$

Greedy Bernstein Algorithm

Given f and we wish to estimate $\int_a^b f(x)dx$ with budget N.

Step 1. **(Initial SD-estimation sample)** Given a region $[a,b]$ with total budget N, set

$$n = \mathrm{round}(0.1N).$$

Draw n uniform samples on $[a,b]$ and store them as

$$\mathcal{S} = \{x_1,\dots,x_n\}, \qquad |\mathcal{S}| = n.$$

(Optionally store $f(x_i)$ alongside each x_i.)

Step 2. **(Compute candidate SDs and split-point selection)** For each candidate $x_i\in\mathcal{S}$, define the induced partitions

$$\mathcal{S}_l(x_i) = \{x\in\mathcal{S} : x\le x_i\}, \qquad \mathcal{S}_r(x_i) = \{x\in\mathcal{S} : x > x_i\}.$$

Compute the empirical standard deviations

$$\hat{\mathrm{sd}}_l(x_i;f) = \hat{\mathrm{sd}}\big(\{f(x) : x\in\mathcal{S}_l(x_i)\}\big), \qquad \hat{\mathrm{sd}}_r(x_i;f) = \hat{\mathrm{sd}}\big(\{f(x) : x\in\mathcal{S}_r(x_i)\}\big).$$

Define the Bernstein split objective

$$h(x_i) := (x_i-a)\cdot\hat{\mathrm{sd}}_l(x_i;f) + (b-x_i)\cdot\hat{\mathrm{sd}}_r(x_i;f).$$

Choose the split point by discrete minimisation:

$$x^* = \arg\min_{x_i\in\mathcal{S}} h(x_i).$$

Step 3. **(Assign samples to children)** Using the chosen split x^*, define the child sample sets

$$\mathcal{S}_l := \{x\in\mathcal{S} : x\le x^*\}, \qquad \mathcal{S}_r := \{x\in\mathcal{S} : x > x^*\}.$$

Then $\mathcal{S}_l\cup\mathcal{S}_r = \mathcal{S}$ and $\mathcal{S}_l\cap\mathcal{S}_r = \emptyset$.

Step 4. **(Allocate remaining budget)** Let the remaining (non-$\mathcal{S}$) budget be

$$N_{\text{rem}} := N - |\mathcal{S}|.$$

Let

$$\hat{\text{sd}}_L^* := \hat{\text{sd}}_l(x^*; f), \qquad \hat{\text{sd}}_R^* := \hat{\text{sd}}_r(x^*; f).$$

Allocate N_{rem} between left and right by the Bernstein allocation rule:

$$N_l = \text{round}\left(N_{\text{rem}} \frac{(x^* - a)\hat{\text{sd}}_L^*}{(x^* - a)\hat{\text{sd}}_L^* + (b - x^*)\hat{\text{sd}}_R^*} \right), \qquad N_r = N_{\text{rem}} - N_l.$$

These N_l and N_r are the **remaining budgets** for new samples in each child region.

Step 5. **(Top up child SD-estimation sets if needed)** For the left child region $[a, x^*]$ with total budget $N_l + |\mathcal{S}_l|$, require an SD-estimation set size

$$n_l = \text{round}(0.1(N_l + |\mathcal{S}_l|)).$$

If $|\mathcal{S}_l| < n_l$, draw additional uniform samples

$$\mathcal{S}_l' \subset [a, x^*], \qquad |\mathcal{S}_l'| = n_l - |\mathcal{S}_l|,$$

and update

$$\mathcal{S}_l \leftarrow \mathcal{S}_l \cup \mathcal{S}_l', \qquad N_l \leftarrow N_l - |\mathcal{S}_l'|.$$

Apply the same procedure to the right child region $[x^*, b]$.

Step 6. **(Recursive splitting)** Repeat Steps 2–5 on $[a, x^*]$ using the updated $(\mathcal{S}_l, N_l)$ and on $[x^*, b]$ using $(\mathcal{S}_r, N_r)$, whenever the region is eligible for further splitting.

Step 7. **(Termination)** The algorithm terminates when, for every region, the total effective budget in that region (remaining budget plus the stored SD-estimation points in that region) is strictly less than $0.1N$:

$$N_{\text{region,rem}} + |\mathcal{S}_{\text{region}}| < 0.1N.$$

Remark 1. *Unlike the Greedy Hoeffding approach, which relies on estimating extrema and thus implicitly requires strong regularity assumptions on f, the Greedy Bernstein method only requires sampling and estimating local variance. The resulting split optimisation is discrete, making it feasible.*

Comparison

Estimating integral of $f(x) = e^x$ on $[0, 10]$

In this case, our integrand f has nice property (convex and increasing). Therefore, our Hoeffding here does not need to require Newton's method for optimisation, and we can get exact minimiser (splitting points).

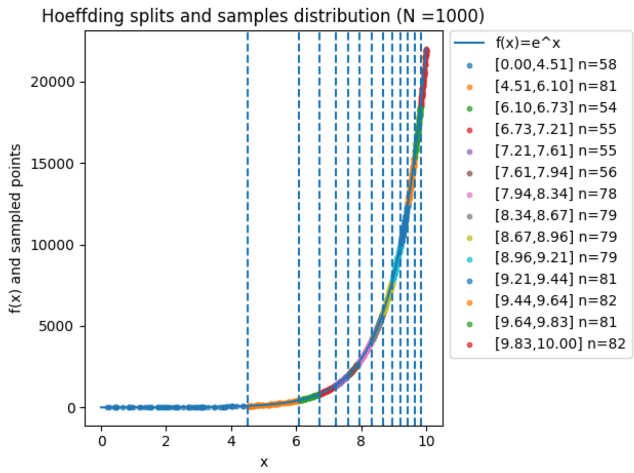

Figure 1: Splits and sample distrition using Greedy Hoeffding algorithm.

As for Greedy Bernstein algorithm, we get

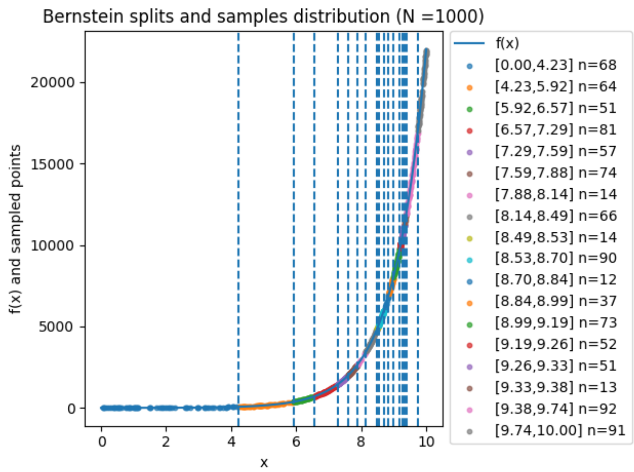

Figure 2: Splits and sample distribution of Greedy Bernstein algorithm.

Now since both estimators are unbiased (because each interval estimate is unbiased), we can compare their variance (also MSE) as N (increases)

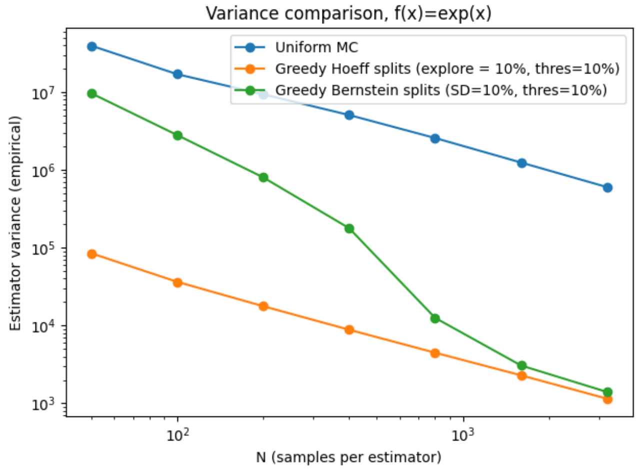

Figure 3: Comparison of uniform Monte Carlo (MC), greedy Hoeffding and greedy Bernstein estimate for $f(x) = e^x$.

Estimating integral of $f(x) = x + \sin(x)$ on $[0, 5]$

Here we would require Newton's for greedy Hoeffding. Below are the results.

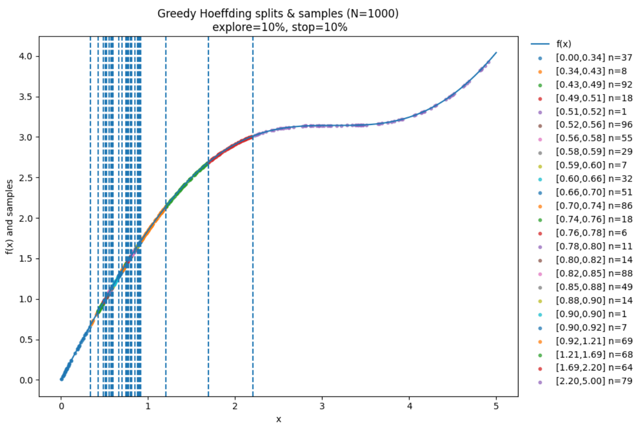

Figure 4: Splits and point distribution of greedy Hoeffding for $f(x) = x + \sin(x)$

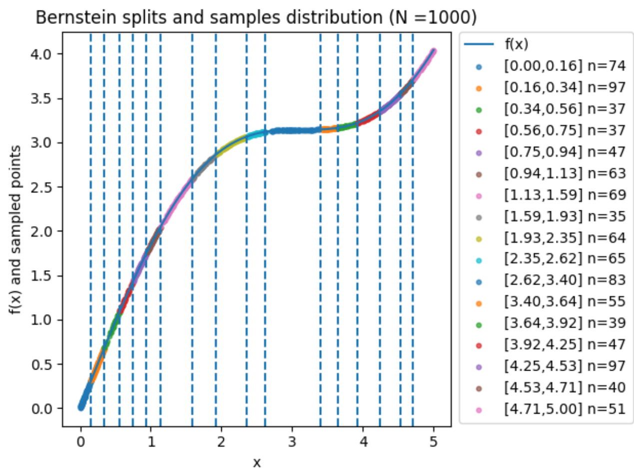

Figure 5: Splits and point distribution of greedy Bernstein for $f(x) = x + \sin(x)$

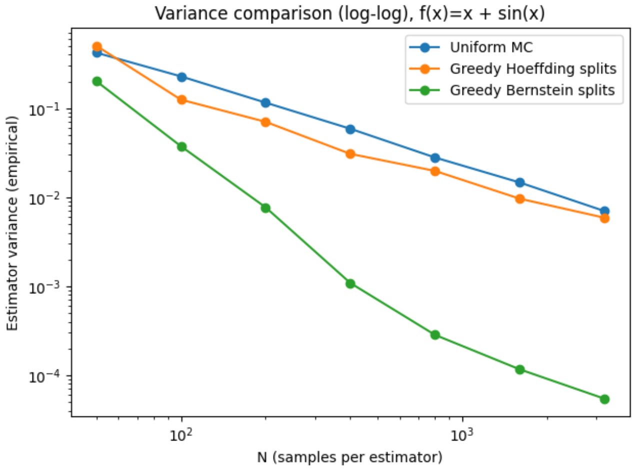

Figure 6: Comparison of uniform Monte Carlo (MC), greedy Hoeffding and greedy Bernstein estimate for $f(x) = x + \sin(x)$.